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Takagi

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(54) **STRUCTURE AND IMAGE FORMING APPARATUS**

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G03G 21/16 (2006.01)

(52) **U.S. Cl.**
CPC ***G03G 21/1619*** (2013.01); ***G03G 21/1647*** (2013.01)

(58) **Field of Classification Search**
CPC G03G 21/1619; G03G 21/1647
See application file for complete search history.

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(57) **ABSTRACT**

Each of four outer side surfaces of each of a plurality of square pipes includes a groove portion formed along a longitudinal direction of each of the plurality of square pipes, and a pair of ridgeline portions on both sides of the groove portion. An additional member has a contact plane that comes in contact with one or both of the pair of ridgeline portions of each of the plurality of target pipes. A plurality of cantilever portions of the additional member come in contact with only one of the pair of ridgeline portions. An overhang portion of the additional member is formed between two specific cantilever portions of the plurality of cantilever portions, and comes in contact with both of the pair of ridgeline portions of the specific pipe. An edge of each of the plurality of cantilever portions is welded to the groove portion.

5 Claims, 4 Drawing Sheets

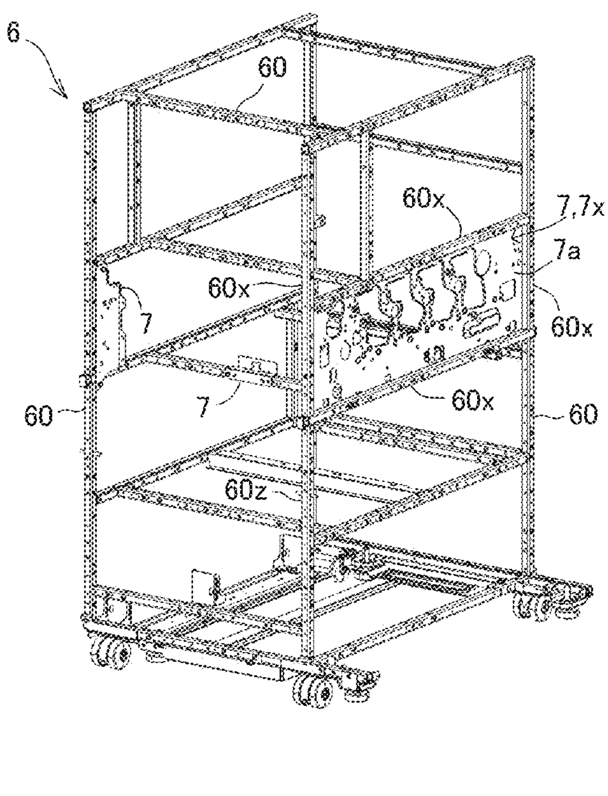
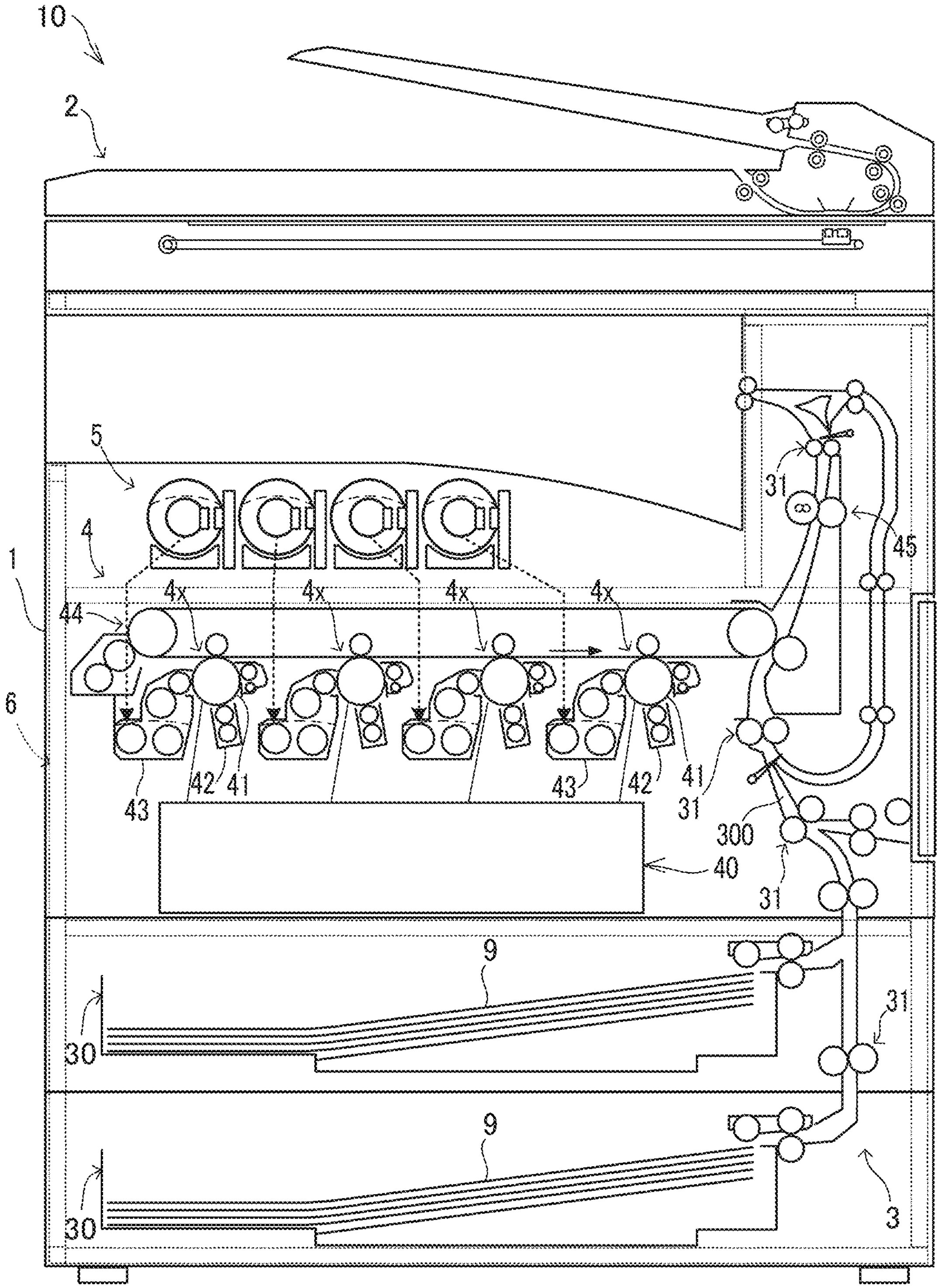


FIG.1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44
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6
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31
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9
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31
9
30
3

FIG.2

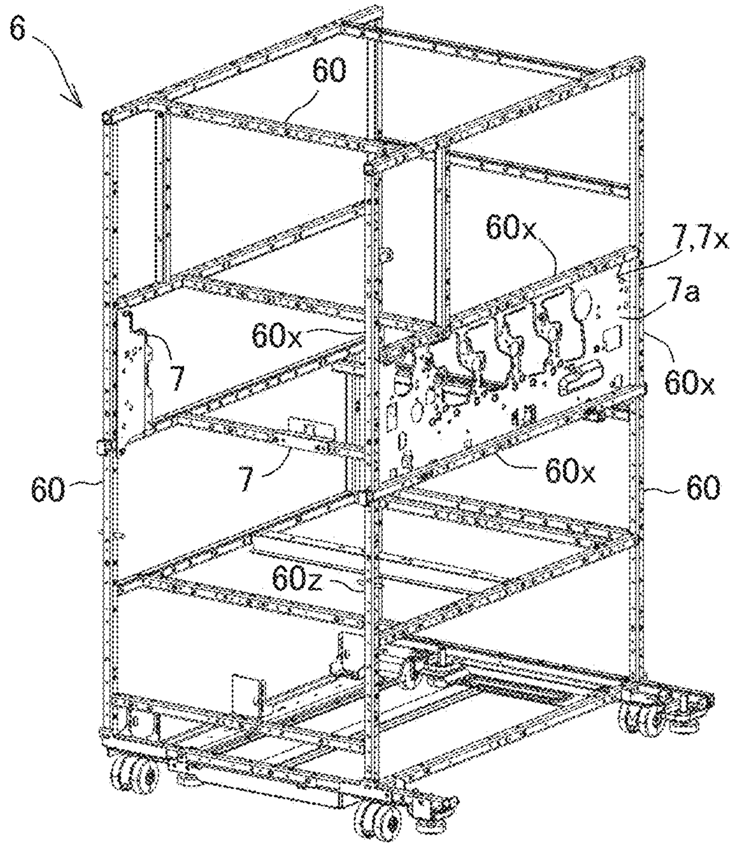
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FIG.3

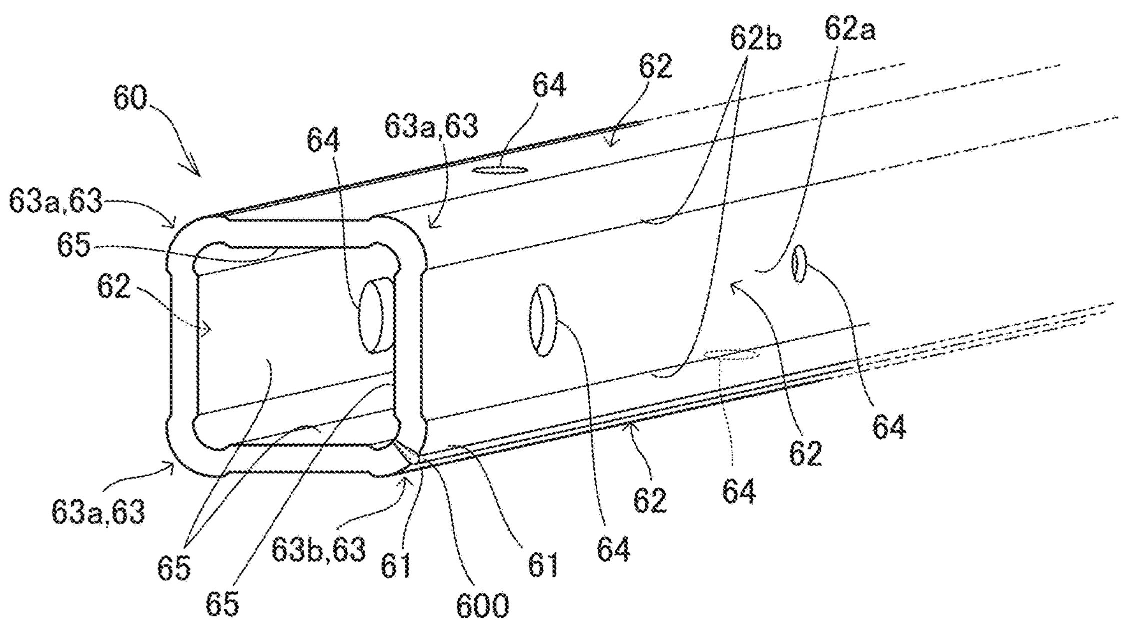
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FIG.4

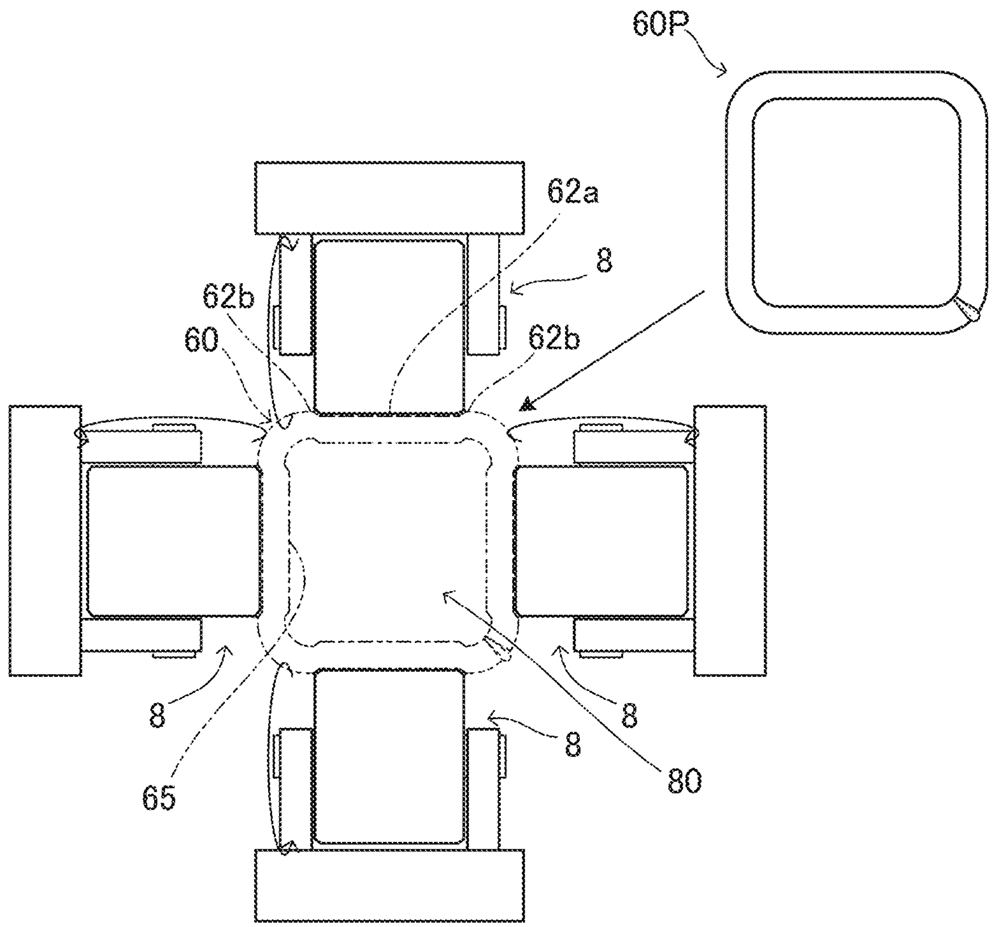
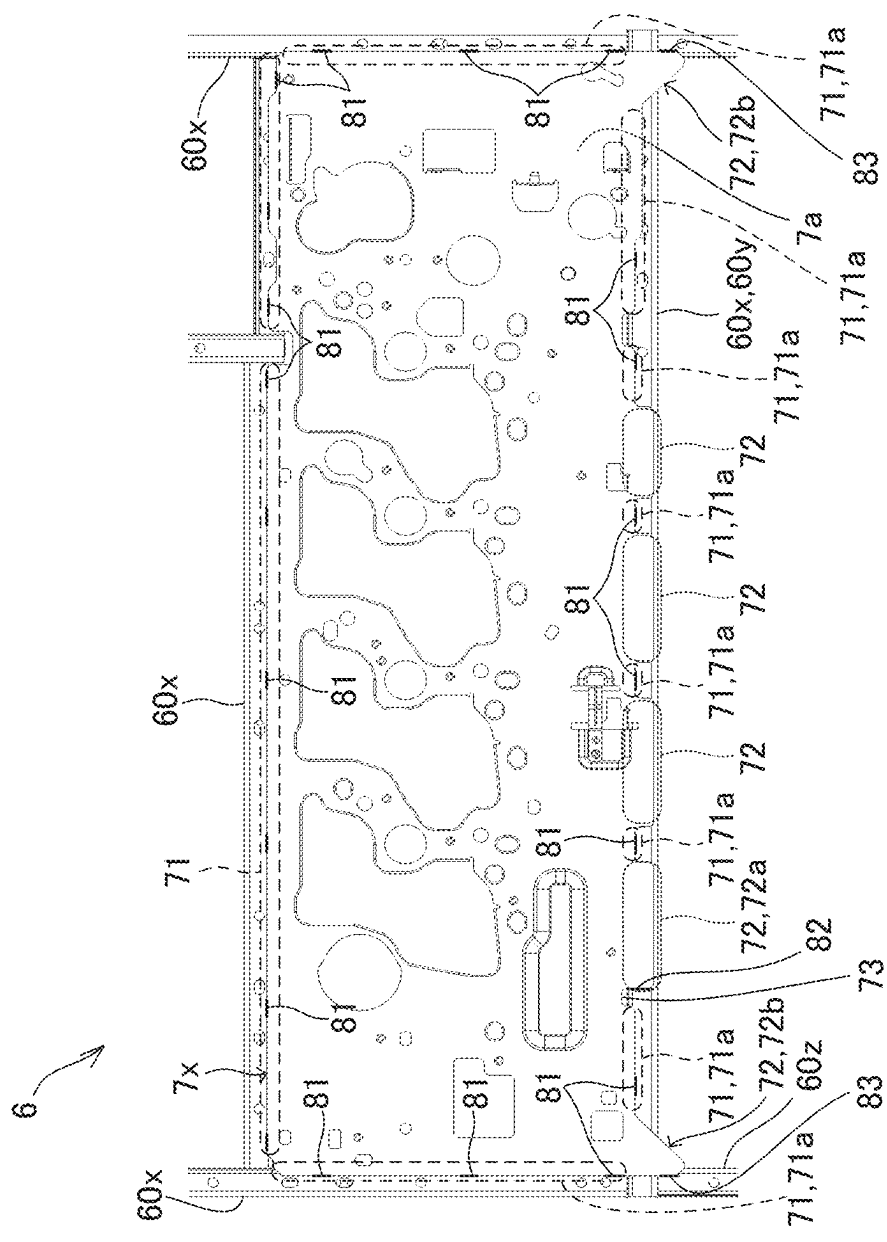


FIG.5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7x
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72,72b
72,72a
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7a
82
73
83
60z

STRUCTURE AND IMAGE FORMING APPARATUS

INCORPORATION BY REFERENCE

This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-094455 filed on Jun. 8, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

The present disclosure relates to a structure including a plurality of square pipes joined together, and an image forming apparatus including the structure.

The image forming apparatus includes a sheet feed portion, a printing portion, a developing agent container, and the like. Furthermore, the image forming apparatus includes a structure that supports the sheet feed portion, the printing portion, and the developing agent container.

For example, a structure is known that includes a plurality of square pipes joined by welding. A structure having such a configuration has high strength.

SUMMARY

A structure according to one aspect of the present disclosure has a plurality of joined square pipes and an additional member attached to some of the plurality of square pipes. Four corners of each of the plurality of square pipes include three first corners formed by bending a single metal plate, and one second corner including two edge portions of the metal plate and a weld portion joining the two edge portions. Each of four outer side surfaces of each of the plurality of square pipes includes a groove portion formed along a longitudinal direction of each of the plurality of square pipes, and a pair of ridgeline portions forming edge portions on both sides of the groove portion. The additional member has a contact plane that is a plane that comes in contact with one or both of the pair of ridgeline portions of each of the plurality of target pipes that are some of the plurality of square pipes. Outer edge portions of the additional member include a plurality of cantilever portions and an overhang portion. The plurality of cantilever portions each include a region that comes in contact with only one of the pair of ridgeline portions of each of the plurality of target pipes in the contact plane. The overhang portion is formed between two specific cantilever portions of the plurality of cantilever portions and protrudes farther outward than the two specific cantilever portions, and includes a region in the contact plane that comes in contact with both of the pair of ridgeline portions of a specific pipe that is one of the plurality of target pipes. An edge of each of the plurality of cantilever portions is welded to the groove portion of each of the plurality of target pipes.

An image forming apparatus according to another aspect of the present disclosure includes the structure, and a printing portion attached to the structure and configured to form an image on a sheet.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a configuration diagram of an image forming apparatus including a structure according to an embodiment.

FIG. **2** is a perspective view of a structure according to an embodiment.

FIG. **3** is a perspective view of one square pipe in a structure according to an embodiment.

FIG. **4** is a diagram showing a part of a manufacturing process for manufacturing a square pipe in a structure according to an embodiment.

FIG. **5** is a front view of a plurality of target pipes and target additional members in a structure according to an embodiment.

DETAILED DESCRIPTION

Embodiments according to the present disclosure will be described below with reference to the drawings. Note that the following embodiments are examples of embodying a technique according to the present disclosure, and do not limit the technical scope of the present disclosure.

[Configuration of Image Forming Apparatus **10**]

A structure **6** according to an embodiment is employed in an image forming apparatus **10**. As shown in FIG. **1**, the image forming apparatus **10** includes a main body unit **1** and an image reading unit **2**.

The main body unit **1** includes a sheet feed portion **30**, a paper conveying device **3**, a printing portion **4**, one or more developing agent containers **5**, and the structure **6**. The image reading unit **2** is connected to an upper portion of the main body unit **1**. The image reading unit **2** is a device that reads an image of a document.

The sheet feed portion **30** is a device that stores a plurality of sheets of paper and feeds out the stored sheets one by one to a paper conveying path **300**. The paper conveying device **3** includes a plurality of conveying roller pairs **31** that convey the paper along the paper conveying path **300**.

The printing portion **4** is a device that forms an image on the paper supplied from the sheet feed portion **30** via the paper conveying device **3**. In the example shown in FIG. **1**, the printing portion **4** forms an image on the paper using an electrophotographic method. The paper is an example of a sheet.

The electrophotographic printing portion **4** includes a laser scanning unit **40**, one or more image forming portions **4*x***, a transfer device **44**, and a fixing device **45**.

In the example shown in FIG. **1**, the printing portion **4** is a tandem type color image printing device. Therefore, the printing portion **4** includes four image forming portions **4*x*** corresponding to toners of four colors: yellow, magenta, cyan, and black. Furthermore, the image forming apparatus **10** includes four developing agent containers **5** corresponding to the four image forming portions **4*x***.

In each of the image forming portions **4*x***, a drum-shaped photoconductor **41** rotates, and a charging device **42** charges the outer peripheral surface of the photoconductor **41**. Furthermore, a laser scanning unit **40** writes an electrostatic latent image on the outer peripheral surface of the photoconductor **41**, and a developing device **43** develops the electrostatic latent image into a toner image.

Furthermore, in the transfer device **44**, an intermediate transfer belt rotates while coming in contact with the four photoconductors **41**, and four primary transfer devices corresponding to the four image forming portions **4***x* transfer the toner images to the intermediate transfer belt. Furthermore, a secondary transfer device transfers the toner image on the intermediate transfer belt to the paper being conveyed via the paper conveying path **300**.

The fixing device **45** fixes the toner image on the paper by heating and pressing the toner image on the paper. The paper conveying device **3** discharges the paper on which the image is formed from the paper conveying path **300**.

Each developing agent container **5** supplies toner to a corresponding developing device **43** in the printing portion **4**. The toner is an example of a developing agent. Note that the printing portion **4** may be a device that forms an image on the paper using an inkjet method or another method.

The sheet feed portion **30**, the paper conveying device **3**, the printing portion **4**, and the developing agent containers **5** are attached to the structure **6**. The image reading unit **2** is connected to an upper portion of the structure **6**.

The structure **6** includes a plurality of joined square pipes **60** (see FIG. **2**). In the structure **6**, a plurality of metal square pipes **60** are joined by welding. Such a structure **6** has high strength.

A typical square pipe is manufactured by bending a single metal plate and welding two edge portions of the metal plate together.

On the other hand, one or more additional members **7** may be attached to some of the plurality of square pipes **60** along outer side surfaces of some of the plurality of square pipes **60** (see FIG. **2**). In this case, the additional members **7** may be required to be attached with high positional accuracy.

For example, the additional members **7** are sheet metal members manufactured by bending and/or drilling metal plate.

In the present embodiment, the structure **6** includes a plurality of additional members **7** attached to some of the plurality of square pipes **60** (see FIG. **2**). The additional members **7** are fixed to a part of the plurality of square pipes **60** by welding or screws.

In a conventional structure, due to insufficient accuracy of an outer surface shape of each of the plurality of conventional square pipes, it was difficult to attach the additional members **7** to some of the plurality of conventional square pipes with high positional accuracy.

In the present embodiment, the structure **6** has a configuration that allows the additional members **7** to be attached to the outer side surfaces of some of the square pipes **60** with high positional accuracy. The configuration will be explained below.

[Configuration of Square Pipes **60**]

Each of the plurality of square pipes **60** has four outer side surfaces **62** and four corners **63** (see FIG. **3**). The four outer side surfaces **62** and the four corners **63** are formed to extend in the longitudinal direction of each square pipe **60**.

The four corners **63** include three first corner portions **63***a* and one second corner portion **63***b*. The three first corner portions **63***a* are three bent portions formed by bending a single metal plate. The second corner portion **63***b* includes two edge portions **61** of the metal plate and a weld portion **600** that joins the two edge portions **61**.

The two edge portions **61** and the weld portion **600** are formed to extend in the longitudinal direction of each square pipe **60**.

Each of the square pipes **60** has a plurality of through holes **64** formed in one or more of the four outer side surfaces **62** (see FIG. **3**). Processing of the plurality of through holes **64** in the plurality of square pipes **60** is performed on the flat metal plate before bending is performed.

Some or all of the plurality of through holes **64** are used for attaching components. For example, the components are directly attached to one of the outer side surfaces **62** of each square pipe **60** by fixing members such as screws inserted into the through holes **64**.

In the present embodiment, the image reading unit **2**, paper conveying device **3**, and printing portion **4** are attached to the structure **6** by the fixing members inserted into the through holes **64**.

Note that two or more of the plurality of square pipes **60** may be the same type of member in order to share components. In this case, in some of the plurality of square pipes **60** of the same type, some of the plurality of through holes **64** may not be used for attaching components.

In each square pipe **60**, none of the four outer side surfaces **62** includes a weld portion **600**. Therefore, in each of the square pipes **60**, it is easy to form one or more through holes **64** in each of the four outer side surfaces **62**.

For example, four or more through holes **64** are formed in four outer side surfaces **62** of at least one of the plurality of square pipes **60** (see FIG. **3**).

In each of the square pipes **60**, two of the four outer side surfaces **62** are formed between the three first corner portions **63***a*. The remaining two of the four outer side surfaces **62** are formed between two of the three first corner portions **63***a* and the second corner portion **63***b*.

By employing square pipes **60** as shown in FIG. **3** in the structure **6**, all four outer side surfaces **62** of each square pipe **60** can be used for attaching parts.

In the present embodiment, the three first corner portions **63***a* form a chamfered shape (see FIG. **3**). In addition, in the second corner portion **63***b*, the two edge portions **61** of the metal plate are bent to form a chamfered shape.

In each of the square pipes **60**, the weld portion **600** may be formed to somewhat protrude from the gap between the two edge portions **61**. Even in such a case, the two edge portions **61** form a chamfered shape, and thus the weld portion **600** does not interfere with the components attached to each of the two outer side surfaces **62** adjacent to the second corner portion **63***b*.

Each of the four outer side surfaces **62** of each of the plurality of square pipes **60** has a groove portion **62***a* and a pair of ridgeline portions **62***b* (see FIGS. **3** and **4**).

The groove portion **62***a* is a concave portion formed along the longitudinal direction of each of the plurality of square pipes **60**. The pair of ridgeline portions **62***b* are portions forming the edges on both sides of the groove portion **62***a*. The pair of ridgeline portions **62***b* are also formed along the longitudinal direction of each of the plurality of square pipes **60**.

The pair of ridgeline portions **62***b* form a pair of crown portions between a pair of step portions and the pair of corner portions **63** on both sides of the groove portion **62***a*. Each of the plurality of square pipes **60** has four groove portions **62***a* and four pairs of ridgeline portions **62***b* corresponding to the four outer side surfaces **62**.

For example, the step portion between each of the grooves **62***a* and the pair of ridgeline portions **62***b* on both sides of each of the grooves **62***a* is about 0.3 mm to 0.8 mm.

In the following description, the square pipe before the groove portions **62***a* and the pair of ridgeline portions **62***b* of each of the four outer side surfaces **62** are formed is referred to as the original square pipe **60**P (see FIG. **4**).

As shown in FIG. **4**, the groove portions **62***a* and the pair of ridgeline portions **62***b* of each of the four outer side surfaces **62** of each of the plurality of square pipes **60** are formed by a roll press in which the original square pipe **60**P is passed between four pressure rollers **8**.

Two of the four pressure rollers **8** and the remaining two are arranged to face each other via a forming space **80** into which the original square pipe **60**P is inserted. The four pressure rollers **8** are each fixed at a predetermined position.

By passing the original square pipe **60**P through the forming space **80**, the four pressure rollers **8** rotate while being in pressure contact with the four outer side surfaces of the original square pipe **60**P. Thus, four groove portions **62***a* corresponding to the surface shapes of the four pressure rollers **8** are formed on the four outer side surfaces of the original square pipe **60**P.

In addition, at the same time as the four groove portions **62***a* are being formed, a pair of ridgeline portions **62***b* are formed on both sides of each of the four groove portions **62***a*.

Moreover, when the roll pressing is performed, the inner-side surfaces of the original square pipe **60**P are not restrained. Therefore, the four inner-side surfaces **65** of each of the plurality of square pipes **60** are formed so as to protrude inward corresponding to the four groove portions **62***a* (see FIGS. **3** and **4**).

Even in a case where the dimensional accuracy of the four outer side surfaces of the original square pipe **60**P is not sufficient, the dimensional accuracy of the four groove portions **62***a* and the four pairs of ridgeline portions **62***b* in the square pipe **60** after molding is high.

On the other hand, each of the additional members **7** has a contact plane **7***a* that is a plane that comes in contact with a part of the plurality of square pipes **60** of the structure **6** (see FIGS. **2** and **5**). Hereinafter, one of the additional members **7** will be referred to as a target additional member **7***x* (see FIGS. **2** and **5**).

The plurality of square pipes **60** include a plurality of target pipes **60***x* to which target additional members **7***x* are attached (see FIGS. **2** and **5**). The contact plane **7***a* of the target additional member **7***x* is a plane that comes in contact with one or both of the pair of ridgeline portions **62***b* in each of the plurality of target pipes **60***x*.

In the example shown in FIG. **5**, the contact plane **7***a* of the target addition member **7***x* comes in contact with one or both of the pair of ridgeline portions **62***b* in each of the four target pipes **60***x*.

In the following description, one of the four target pipes **60***x* will be referred to as a specific pipe **60***y* (see FIG. **5**). In the example shown in FIG. **5**, the specific pipe **60***y* is arranged along a lower edge portion of the target additional member **7***x*.

An outer edge portion of the target additional member **7***x* includes a plurality of cantilever portions **71** and at least one overhang portion **72** (see FIG. **5**). In the example shown in FIG. **5**, the outer edge portion of the target additional member **7***x* includes six overhang portions **72**.

Each of the plurality of cantilever portions **71** is a portion that includes a region that comes in contact with only one of the pair of ridgeline portions **62***b* of each of the plurality of target pipes **60***x* on the contact plane **7***a* of the target additional member **7***x*.

The edges of each of the plurality of cantilever portions **71** are welded to the groove portions **62***a* of each of the plurality of target pipes **60***x*. The edges of each of the plurality of cantilever portions **71** are welded to the groove portions **62***a* by one or more first weld portions **81**. A first weld portion **81** is formed along the longitudinal direction of each target pipe **60***x*.

When the contact plane **7***a* is in contact with at least one of the pair of ridgeline portions **62***b* in each of the plurality of target pipes **60***x*, a slight gap is formed between the target additional member **7***x* and the groove portion **62***a*. There is no particular problem in welding two members in a case where the members are placed with a gap of about 0.3 mm to 0.8 mm between them.

On the other hand, each of the overhang portions **72** is formed between two specific cantilever portions **71***a* of the plurality of cantilever portions **71** so as to protrude further outward than the two specific cantilever portions **71***a*. Each of the overhang portions **72** is a portion including a region that comes in contact with both of the pair of the ridgeline portions **62***b* of the specific pipe **60***y* on the contact plane **7***a* of the target additional member **7***x*.

Two specific cantilever portions **71***a* on both sides of each overhang portions **72** are fixed by the first weld portions **81**. Thus, even when each of the overhang portions **72** is not fixed to the specific pipe **60***y*, the target additional member **7***x* is firmly fixed to the plurality of target pipes **60***x*.

In the present embodiment, an edge of a first overhang portion **72***a*, which is one of the plurality of overhang portions **72**, is welded to the groove portion **62***a* of the specific pipe **60***y* by the second weld portion **82**. The second weld portion **82** is formed along a direction intersecting the pair of ridgeline portions **62***b* of the specific pipe **60***y*.

In a case where the first overhang portion **72***a* and the two specific cantilever portions **71***a* on both sides of the first overhang portion **72***a* are welded to the specific pipe **60***y*, the first weld portion **81** and the second weld portion **82** are formed relatively close to each other. In this case, there is a possibility that a relatively large stress will be applied to the first overhang portion **72***a* due to a dimensional error in the height of the pair of ridgeline portions **62***b* of the specific pipe **60***y*.

Therefore, the target additional member **7***x* has a notch **73** formed between the first overhang portion **72***a* and one of the two specific cantilever portions **71***a* on both sides of the first overhang portion **72***a* The notch **73** is formed to be recessed inward of the target additional member **7***x* to a position where the pair of ridgeline portions **62***b* of the specific pipe **60***y* are exposed.

That is, a portion of the outer edge portion of the target additional member **7***x* between the first overhang portion **72***a* and one of the two specific cantilever portions **71***a* on both sides of the first overhang portion **72***a* does not come in contact with the specific pipe **60***y*.

Fixing the first overhang portion **72***a* by the second weld portion **82** is suitable when it is necessary to fix the target additional member **7***x* more firmly. The notch **73** has an effect of relieving stress applied to the first overhang portion **72***a* fixed by the second welding portion **82**.

In the present embodiment, the plurality of overhang portions **72** include one or more second overhang portions **72***b* formed to extend to the position of an adjacent pipe **60***z* adjacent to the specific pipe **60***y*. The adjacent pipe **60***z* is one of the plurality of square pipes **60** other than the plurality of target pipes **60***x*. In the example shown in FIG. **5**, the plurality of overhang portions **72** include two second overhang portions **72***b*.

The contact plane **7***a* of each second overhang portion **72***b* comes in contact with both of the pair of ridgeline portions **62***b* of the specific pipe **60***y* and one of the pair of ridgeline portions **62***b* of the adjacent pipe **60***z*.

The edge of each of the second overhang portions **72***b* is welded to the groove portion **62***a* of the adjacent pipe **60***z* by a third weld portion **83**. The third weld portion **83** is formed along the longitudinal direction of the adjacent pipe **60***z*.

The third weld portion **83** is formed at a position sufficiently spaced from the second weld portion **82** formed on the two specific cantilever portions **71***a* on both sides of each of the second overhang portions **72***b*. Therefore, the stress applied to the second overhang portion **72***b* due to a dimensional error in the heights of the pair of ridgeline portions **62***b* of the specific pipe **60***y* and the adjacent pipe **60***z* is not large.

Note that a notch **73** may be formed between each of the second overhang portions **72***b* and one of the two specific cantilever portions **71***a* on both sides of each of the second overhang portions **72***b*.

In the present embodiment, of the plurality of overhang portions **72**, the portions other than the first overhang portions **72***a* and the second overhang portions **72***b* are not welded to any of the plurality of square pipes **60**.

The target additional member **7***x* is positioned with high accuracy in a state where the contact plane **7***a* of the target additional member **7***x* is in contact with one or both of the pair of ridgeline portions **62***b* of each of the plurality of target pipes **60***x*.

By employing the structure **6**, it is possible to weld the target additional member **7***x* to the outer side surfaces **62** of the plurality of target pipes **60***x* with high positional accuracy.

SUPPLEMENTARY NOTES

Hereinafter, a summary of the disclosure extracted from the above-described embodiments will be added. Note that each configuration and each processing function described in the following supplementary notes can be selected and combined as desired.

Supplementary Note 1

A structure including a plurality of joined square pipes and an additional member attached to some of the plurality of square pipes, wherein

four corners of each of the plurality of square pipes include three first corners formed by bending a single metal plate, and one second corner including two edge portions of the metal plate and a weld portion joining the two edge portions;

each of four outer side surfaces of each of the plurality of square pipes includes a groove portion formed along a longitudinal direction of each of the plurality of square pipes, and a pair of ridgeline portions forming edge portions on both sides of the groove portion;

the additional member has a contact plane that is a plane that comes in contact with one or both of the pair of ridgeline portions of each of the plurality of target pipes that are some of the plurality of square pipes;

outer edge portions of the additional member include

a plurality of cantilever portions, each including a region that comes in contact with only one of the pair of ridgeline portions of each of the plurality of target pipes in the contact plane, and

an overhang portion formed between two specific cantilever portions of the plurality of cantilever portions and protruding farther outward than the two specific cantilever portions, and including a region in the contact plane that comes in contact with both of the pair of ridgeline portions of a specific pipe that is one of the plurality of target pipes; and

an edge of each of the plurality of cantilever portions is welded to the groove portion of each of the plurality of target pipes.

Supplementary Note 2

The structure according to supplementary note 1, wherein

an edge of the overhang portion of the additional member is welded to the groove portion of the specific pipe, and

the additional member has a notch formed to be recessed inward between the overhang portion and one of the two specific cantilever portions on both sides of the overhang portion to a position where the pair of ridgeline portions of the specific pipe are exposed.

Supplementary Note 3

The structure according to supplementary note 1, wherein

the contact plane of the overhang portion of the additional member comes in contact with the pair of ridgeline portions of the specific pipe and one of the pair of ridgeline portions of an adjacent pipe adjacent to the specific pipe of the plurality of square pipes, and

an edge of the overhang portion is welded to the groove portion of the adjacent pipe.

Supplementary Note 4

The structure according to any one of supplementary notes 1 to 3, wherein

the groove portion and the pair of ridgeline portions on each of the four outer side surfaces of each of the plurality of square pipes are formed by a roll press in which an original square pipe before the groove portion and the pair of ridgeline portions of each of the four outer side surfaces are formed is passed between four pressure rollers that press against four outer side surfaces of the original square pipe.

Supplementary Note 5

An image forming apparatus including

the structure described in any one of supplementary notes 1 to 4, and

a printing portion attached to the structure and configured to form an image on a sheet.

It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

The invention claimed is:

1. A structure comprising a plurality of joined square pipes and an additional member attached to some of the plurality of square pipes, wherein

four corners of each of the plurality of square pipes comprise three first corners formed by bending a single metal plate, and one second corner including two edge portions of the metal plate and a weld portion joining the two edge portions;

each of four outer side surfaces of each of the plurality of square pipes includes a groove portion formed along a longitudinal direction of each of the plurality of square pipes, and a pair of ridgeline portions forming edge portions on both sides of the groove portion;

the additional member has a contact plane that is a plane that comes in contact with one or both of the pair of ridgeline portions of each of the plurality of target pipes that are some of the plurality of square pipes;

outer edge portions of the additional member include

a plurality of cantilever portions, each including a region that comes in contact with only one of the pair of ridgeline portions of each of the plurality of target pipes in the contact plane, and

an overhang portion formed between two specific cantilever portions of the plurality of cantilever portions and protruding farther outward than the two specific cantilever portions, and including a region in the contact plane that comes in contact with both of the pair of ridgeline portions of a specific pipe that is one of the plurality of target pipes; and

an edge of each of the plurality of cantilever portions is welded to the groove portion of each of the plurality of target pipes.

2. The structure according to claim **1**, wherein

an edge of the overhang portion of the additional member is welded to the groove portion of the specific pipe, and

the additional member has a notch formed to be recessed inward between the overhang portion and one of the two specific cantilever portions on both sides of the overhang portion to a position where the pair of ridgeline portions of the specific pipe are exposed.

3. The structure according to claim **1**, wherein

the contact plane of the overhang portion of the additional member comes in contact with the pair of ridgeline portions of the specific pipe and one of the pair of ridgeline portions of an adjacent pipe adjacent to the specific pipe of the plurality of square pipes, and

an edge of the overhang portion is welded to the groove portion of the adjacent pipe.

4. The structure according to claim **1**, wherein

the groove portion and the pair of ridgeline portions on each of the four outer side surfaces of each of the plurality of square pipes are formed by a roll press in which an original square pipe before the groove portion and the pair of ridgeline portions of each of the four outer side surfaces are formed is passed between four pressure rollers that press against four outer side surfaces of the original square pipe.

5. An image forming apparatus, comprising

the structure according to claim **1**, and

a printing portion attached to the structure and configured to form an image on a sheet.

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